

Systematic Literature Review Of Efficacy & Safety Of Sitagliptin In Elderly Type 2 Diabetes Mellitus Patients

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Introduction

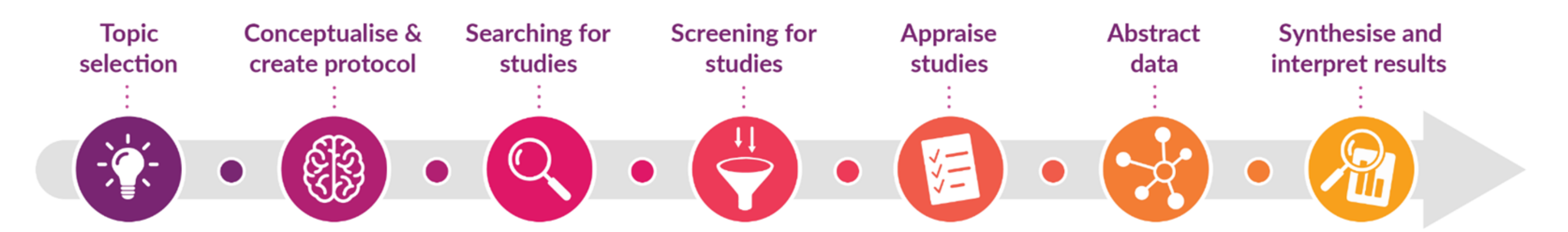
- India’s population trajectory between the years 2000 and 2050 indicates a 55% increase in the total population, with a 326% increase of 60+ age group (76 to 324 million) and a 700% increase in 80+ age group individuals (6 to 48 million).¹
- According to Longitudinal Ageing Study in India (LASI), the self-reported prevalence of diabetes mellitus among Indian adults aged 60 and above was 14%.²
- The presence of comorbidities, increased predisposition to hypoglycaemic events, individual care needs, and lack of resilience leads to an increased risk of frailty, making disease management more complex in older adults.
- Dipeptidyl peptidase IV (DPP-4) inhibitors have an intermediate effect on patients with T2DM and are also less likely to cause weight gain, hypoglycemia and fewer side effects than other OADs and insulin.^{3,4,5}

Objective

To examine the evidence of efficacy and safety of sitagliptin, DPP-4 inhibitor in elderly patients with type 2 diabetes mellitus (T2DM).

Methods

A detailed literature search was conducted using relevant terms in PubMed to identify articles published in the last ten years.



Results

PARAMETERS	STREAM ⁶	STRATEGY ³	CompoSIT-R ⁷	ASSET-K study (Elderly) ⁸	JAMP study (Elderly) ⁹
Subjects	176	681	614	831	651
Age	65–80 years (Mean 70.6 ± 3.9 and 71.9 ± 4.4)	≥65 years (Mean 68.4 ± 2.8)	≥25 years (Mean 67.7 ± 8.5 and 66.6 ± 8.6)	<65 years 65–74 years ≥75 years	<65 years (332) Mean 54.5 ± 7.4 65–74 years (189) Mean 69.2 ± 2.9 ≥75 years (130) Mean 80.0 ± 4.7
Duration in weeks	52	16	24	104	52
HbA1c Baseline Range	7.4–10.4%	≥7.0% & ≤10.0%	≥7.0% & ≤9.5%	-	≥6.9%
Mean HbA1c Baseline	7.8 ± 0.9 Vs 7.6 ± 0.5	8.0 ± 0.8	7.7 ± 0.7 Vs 7.8 ± 0.7	8.1 ±1.2% 7.6 ± 0.9 % 7.7 ± 0.9 %	7.62 ± 0.89 (n = 318); >65 years
Treatment	Sitagliptin /Sitagliptin+ additive anti-diabetic drug & control group	Metformin & sitagliptin for 16 weeks	Metformin±SU with Sitagliptin or Dapagliflozin	Sitagliptin for 2 years	Sitagliptin
HbA1c Target	< 7.4% at week 48	<7% and <7.5% at week 16	<7% at week 24	<6.9 %	-
Achieved the target	< 7.4% (69.5% vs 47.2%) at 12 weeks < 7.4% (69.4% vs. 52.2%) at 24 weeks < 7.4% (64.8% vs. 53.8%) at 48 weeks	HbA1c <7% (44.9%) HbA1c <7.5% (67.2%)	<7% 43% vs 27%)	<65 years 7.4 ± 1.0 %, (12 months) and 7.3 ± 0.9 % (24 months) 65-74 years and ≥75 years 7.0 ± 0.7 % and 7.1 ± 0.7 %; at 12 and 24 months	6.94 ± 0.75 (n = 311) At 3 months 6.96 ± 0.89 (n = 287) At 12 months
Safety	AE recorded in 9.5% and 6.8% with hypoglycemia in one case in both groups	4.1 and 4.3% without occurrence of severe hypoglycemia	48.9% vs 51.6% experienced one or more AEs 7.8% vs 13.7% drug related hypoglycemia	Constipation (8), rash (5), diarrhea (2), edema (2), and nausea (2) Hypoglycemia 5.73% (<65 years), 5.79% (65–74 years), & 3.90 events per 100 patients years in those aged ≥75 years	Hypoglycemia (0.3%)

Effect Of Sitagliptin On HbA1C

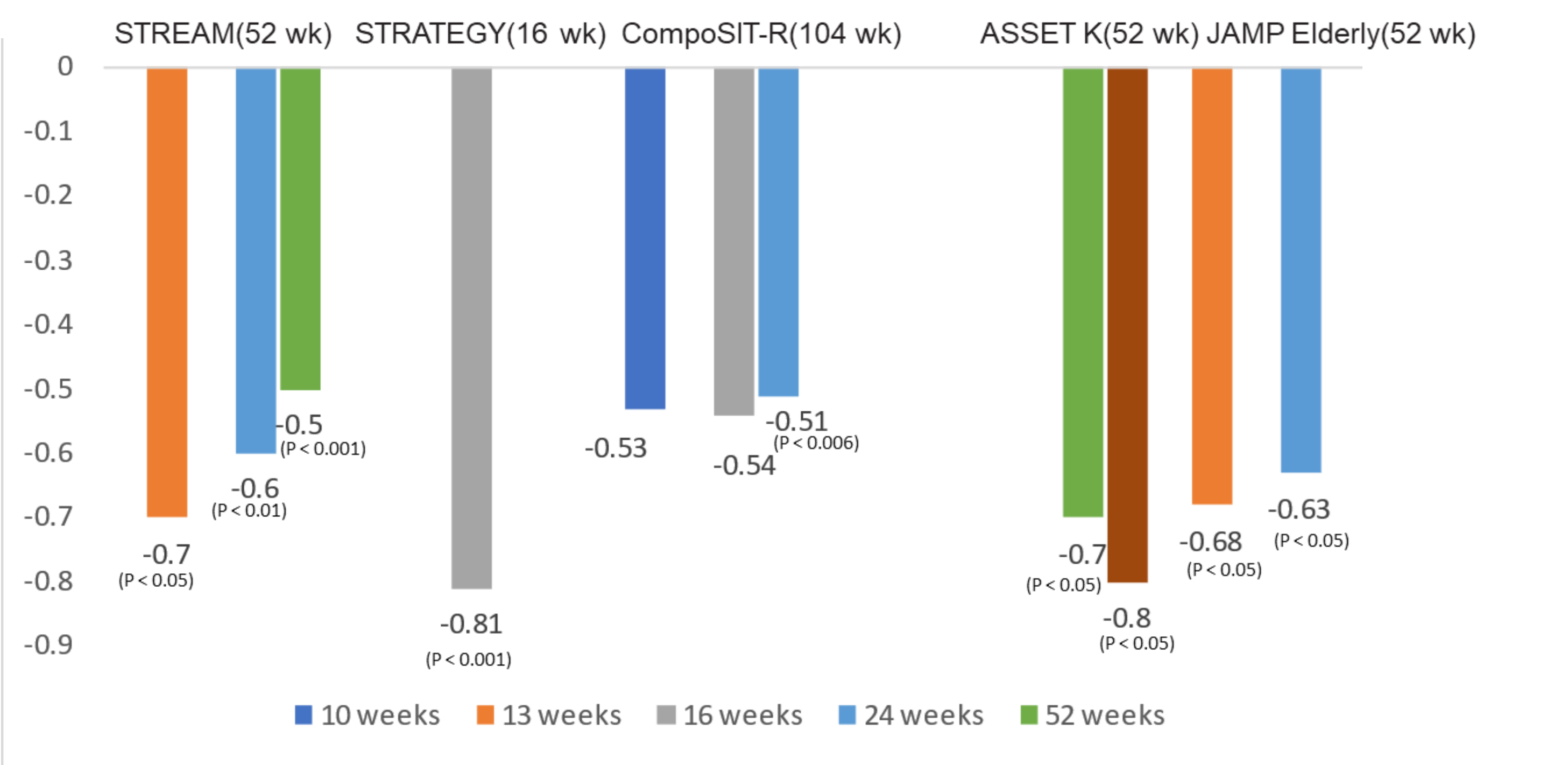
- The Stream study (Nagao et al 2023) in elderly T2DM patients reported a significant reduction (p=0.0001) in fasting blood glucose (−27.2 mg/dL) and HbA1c (−0.61%) at 12 months.⁴
- In The CompoSIT-R study (R Scott et al 2018), at week 24, 43% of elderly patients with T2DM inadequately controlled on metformin+sulfonylurea having mild renal insufficiency, achieved target HbA1c of <7% in sitagliptin group.⁵

Conclusion

- Sitagliptin provides clinically meaningful glycaemic control without increasing the risk of hypoglycaemia.
- Beta cell preservation and reduction in insulin resistance were substantiated by long-term durable efficacy in reducing HbA1c, FBG, & PPBG in elderly T2DM patients.
- These characteristics along with cardio-renal benefits and weight neutrality make sitagliptin as effective treatment of choice for elderly patients.

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Change In HbA1c From Baseline



Effect Of Sitagliptin On Hypoglycemia

In The Strategy study (2020) in elderly Chinese patients metformin/sitagliptin-based dual and triple oral therapy can provide clinically meaningful glycemc control (HbA1c reduction by -0.81% and patients achieving HbA1c <7% and <7.5% as 45% and 67%, respectively) at 16 weeks along with the low incidence of hypoglycaemia.¹

Beta Cell Preservation

	Placebo (BL)	Sitagliptin (BL)	Placebo (18wk)	Sitagliptin (18wk)
HOMA-β	32±27.6	33.0±36.3	36.0±44.3	42±47.9
HOMA-IR	4.4±3.9	4.3±3.5	5.1±5.3	4.1±4.1
Insulinogenic Index	0.27±0.33	0.23±0.22	0.21±0.17	0.31±0.31
Disposition Index	1.0±1.2	0.8±0.5	0.9±0.7	1.3±1.2

Sitagliptin improved 2-h postprandial insulin, 2-h postprandial C-peptide, the insulinogenic index, and the disposition index together with improvement in HOMA-β and reductions in HOMA IR values.¹⁰

Durability

Sitagliptin has a durable effect and stabilizes microvascular complication progression in elderly patients. HbA1c was significantly reduced after 6 months by 0.8% and was maintained for 48 months.¹¹

CV Renal Safety

Sitagliptin did not increase the risk of serious hypoglycaemia and was neutral with respect to CV outcomes over ~3 years of follow-up, in elderly patients with well-controlled diabetes.¹²

Sitagliptin appears to be effective in reducing proteinuria, ameliorating renal function, and producing an anti-inflammatory effect in patients with early-stage diabetic nephropathy.¹⁴

Epidemiological evidence has also showed that elderly T2DM patients treated with DPP-4 inhibitors have better parameters of sarcopenia compared with other glucose lowering drugs.¹⁵